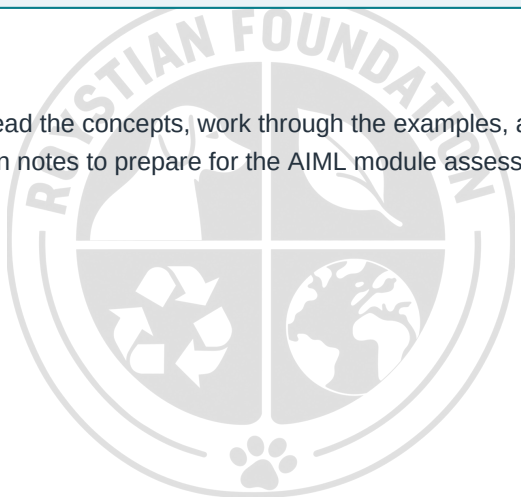
ARTIFICIAL INTELLIGENCE
& MACHINE LEARNING

Learning Guide

Concepts • Practical Examples • Assessment Preparation

10 MODULES
One complete domain guide



A self-paced guide for learners. Read the concepts, work through the examples, and use the practice questions and revision notes to prepare for the AIML module assessments.

Contents

MODULE	CHAPTER
01	Introduction to Artificial Intelligence
02	Machine Learning Fundamentals
03	Data & Model Concepts
04	Real-World AI Applications
05	Model Evaluation & Performance
06	Overfitting, Underfitting & Generalisation
07	Core Algorithms in Practice
08	Neural Networks Explained Simply
09	Responsible & Ethical AI
10	From Notebook to Deployment
—	Practice Assessment and Answer Key

How to use this guide

Each chapter follows a consistent pattern: key ideas, plain-language explanations, examples, and a short checkpoint. The final section contains practice questions with answers. The practice questions are for learning and are not represented as the live platform assessment.

Assessment reminder: the supplied module descriptions state that each module assessment requires 70% to pass and may be retaken as many times as needed. Follow the instructions shown on the live platform for the official attempt.

Module 01: Introduction to Artificial Intelligence

AI is the broad field of building systems that perform tasks commonly associated with human intelligence, such as perception, reasoning, language processing, and decision support.

Learning objectives

- Define AI and distinguish it from fixed-rule automation.
- Explain the relationship between AI, machine learning, and deep learning.
- Recognize narrow AI and three common categories of AI systems.

Artificial Intelligence

AI systems accept inputs, process information, and produce outputs such as classifications, predictions, recommendations, or generated responses. AI is a field rather than one specific algorithm. AI does not automatically imply consciousness, emotion, or human-like understanding.

Automation versus AI

Automation executes a process with reduced manual effort and may use explicit, predefined rules. For example, sending a reminder every Friday is automation. A model that learns from labeled messages to predict spam is an AI/ML task. An automated workflow can include an AI model, but not every automated system uses AI.

Machine Learning

Machine learning (ML) is a subset of AI in which a system learns patterns from data instead of relying entirely on hand-written rules. During training, an algorithm adjusts a model to improve a task. A trained model can then make predictions on new inputs.

Deep Learning

Deep learning (DL) is a subset of ML that uses neural networks with multiple layers to learn representations from data. It is common in image recognition, speech processing, language tasks, and generative systems. Not every AI system uses deep learning.

Narrow AI

Narrow AI is designed for one specific task or a limited set of related tasks. Examples include spam filters, image classifiers, product recommenders, and speech recognition systems. A system may be sophisticated yet remain narrow because its ability does not automatically transfer to unrelated tasks.

Three practical AI categories

Perception systems interpret images, video, or audio—for example, detecting pedestrians. Prediction and recommendation systems estimate outcomes or rank options—for example, detecting unusual transactions or recommending films. Language systems interpret or generate human language—for example, translation, summarization, or chatbots.

Practical scenario

A university uses a scheduled script to send welcome emails, a machine-learning model to rank urgent support tickets, and a deep neural network to detect defects in equipment photos. The email script is rule-based automation; the ticket ranker is ML/AI; and the image model is deep learning, which is also ML and AI.

Checkpoint

AI is the broad field; ML is a subset of AI; DL is a subset of ML. Automation can exist without AI. Narrow AI has task-specific capabilities.

Module 02: Machine Learning Fundamentals

Machine learning methods differ in the feedback or information available during learning. The three major paradigms introduced here are supervised learning, unsupervised learning, and reinforcement learning.

Learning objectives

- Explain the three learning paradigms.
- Describe prediction, loss, and parameter updates in a training loop.
- Choose a learning paradigm for a business problem.

Supervised learning

Supervised learning uses labeled examples: each input is paired with a known target. Classification predicts a category, such as spam/not spam; regression predicts a numeric value, such as delivery time. A model learns a mapping from inputs to targets and is evaluated on data it did not train on.

Unsupervised learning

Unsupervised learning finds structure in data without target labels. Clustering groups similar records, while dimensionality-reduction methods summarize data using fewer variables. For example, a retailer may group customers by purchasing patterns before designing campaigns. The groups require interpretation and are not automatically meaningful.

Reinforcement learning

Reinforcement learning involves an agent interacting with an environment. The agent takes actions and receives rewards or penalties, learning a policy that aims to maximize cumulative reward. It can be used in robotics, game-playing, and some resource-allocation problems. The reward must be designed carefully so the agent does not optimize an unintended behavior.

Loss function

A loss function measures the discrepancy between a model's prediction and a target or objective. Training aims to reduce an appropriate loss. For example, mean squared error measures average squared differences for a regression task. Different tasks require different loss functions; loss is a training signal, not the same thing as every business cost.

The training loop

A simplified loop is: feed training inputs to the model; compute predictions; calculate loss; compute gradients or another update signal; update model parameters; repeat. Validation data helps compare settings and monitor generalization. Training ends according to a chosen stopping rule, and the test set is reserved for final evaluation.

Practical scenario

A bank has historical transactions labeled fraudulent or legitimate and wants to classify new transactions: supervised learning is a natural starting point. A retailer wants to group customers without predefined group labels: unsupervised learning may fit. A robot learns movements from rewards for reaching a target: reinforcement learning may fit.

Checkpoint

Ask what feedback is available: labeled target → supervised; no target labels and looking for structure → unsupervised; actions and rewards over time → reinforcement learning.

Module 03: Data & Model Concepts

Model quality depends heavily on the quality, relevance, and handling of data. This chapter explains features, dataset splits, preprocessing, and data leakage.

Learning objectives

- Explain features, training sets, validation sets, and test sets.
- Describe cleaning, encoding, and scaling.
- Explain how data leakage can make evaluation misleading.

Features and targets

A feature is an input variable used by a model, such as age, transaction amount, or image pixels. A target or label is the value the model is asked to predict in supervised learning. Features should be available at prediction time and should be relevant to the intended task.

Cleaning data

Data cleaning addresses missing values, duplicates, inconsistent units, invalid records, and obvious errors. Choices should be documented because deleting or filling records can change the dataset. Missingness may itself contain information, so the treatment should be based on context rather than a universal rule.

Encoding and scaling

Encoding converts categorical values into numeric representations a model can use; one-hot encoding is one common option for nominal categories. Scaling transforms numeric variables to comparable ranges, for example by standardization or min–max scaling. Not every model needs scaling, and the method should match the algorithm and data.

Training, validation, and test sets

The training set is used to fit model parameters. The validation set supports model selection and hyperparameter tuning. The test set is held back until final evaluation to estimate performance on unseen data. For time-dependent data, random splitting may be inappropriate; splits should reflect the intended future-use setting.

Data leakage

Data leakage occurs when information that would not legitimately be available at prediction time influences training or evaluation. Examples include using a future outcome as a feature, fitting preprocessing on the entire dataset before splitting, or allowing near-duplicate records to appear across splits. Leakage can produce impressive scores that fail in production.

A safe preprocessing workflow

Split data first when appropriate; fit imputation, encoding, and scaling steps on training data only; apply the learned transformations to validation and test data; use validation data for model choices; evaluate the chosen pipeline on the untouched test set. In practice, a pipeline helps keep these steps consistent.

Practical scenario

A model predicts whether a patient will return to a clinic. If a feature records a follow-up diagnosis made after the prediction date, the model is using future information. Its test score may look excellent but is not a valid estimate of what the system could know when making a real prediction.

Checkpoint

Keep the test set untouched until final evaluation. Ask of every feature: would this value genuinely be available at the time the prediction is made?

Module 04: Real-World AI Applications

AI creates value when a suitable task, useful data, measurable objectives, and an appropriate deployment process come together. AI is not the right solution for every problem.

Learning objectives

- Describe use cases in healthcare, finance, retail, and agriculture.
- Identify likely data requirements.
- Recognize cases where simpler non-AI approaches are preferable.

Recommendation systems

A recommendation system ranks items for a user using signals such as previous behavior, item characteristics, and patterns across similar users. Applications include films, music, products, and learning content. Common challenges include cold-start users or items, feedback loops, privacy, and over-personalization.

Computer vision

Computer vision extracts information from images or video. Tasks include image classification, object detection, segmentation, and document reading. Applications include quality inspection, crop monitoring, medical-image support, and traffic analysis. Image quality, labeling, lighting, camera placement, and population coverage affect reliability.

Natural Language Processing

Natural Language Processing (NLP) concerns processing human language. Tasks include sentiment classification, translation, information extraction, summarization, and conversational interfaces. Text can be ambiguous and context-dependent; language systems may produce plausible but incorrect outputs, so important uses require validation.

Predictive maintenance

Predictive maintenance uses equipment sensor history and maintenance records to estimate the likelihood or timing of failures. Useful inputs may include vibration, temperature, operating hours, error codes, and past failures. The goal is to schedule service before a breakdown while avoiding unnecessary maintenance.

Sector examples and data needs

Healthcare may use images, clinical records, and sensor data to support clinical workflows; finance may use transactions and account behavior to flag suspicious activity; retail may use purchases and product interactions for recommendations or demand forecasting; agriculture may use crop images, weather, soil readings, and satellite imagery. Data access, quality, consent, and domain expertise are essential.

When AI may be the wrong tool

A deterministic rule, spreadsheet, database query, or simple statistical method may be cheaper, easier to explain, and more reliable when requirements are stable and explicit. AI may be unsuitable when there is insufficient representative data, the cost of errors is unacceptable without safeguards, or no measurable benefit justifies the complexity.

Practical scenario

A farm wants to identify leaf disease from images. A vision model could help if the photos are representative and labels are reliable. If the task is simply to remind workers to irrigate at a fixed time, a scheduled rule may be a better solution than an AI model.

Checkpoint

Start with the problem and success measure, then assess data, risks, baseline methods, cost, and monitoring needs before choosing AI.

Module 05: Model Evaluation & Performance

The supplied module list skips Module 5. This chapter fills the sequence with foundational evaluation concepts that support the other AIML modules. Confirm the exact Module 5 title and assessment topics if the platform has a different syllabus.

Learning objectives

- Explain why evaluation must use unseen data.
- Recognize common classification metrics and regression error measures.
- Choose a metric that reflects the task's risks.

Evaluation and generalization

Evaluation estimates how well a trained model performs on data it has not seen during fitting. A high training score alone does not show that the model will work in the real world. Use a relevant validation strategy and keep the final test set separate from model selection.

Classification metrics

Accuracy is the proportion of predictions that are correct. Precision is the proportion of predicted positives that are truly positive. Recall is the proportion of actual positives found by the model. The F1 score combines precision and recall using their harmonic mean. A confusion matrix records true positives, true negatives, false positives, and false negatives.

Why accuracy can mislead

If only 1% of transactions are fraudulent, a model that predicts every transaction as legitimate can achieve 99% accuracy while finding no fraud. Precision, recall, class-specific results, and the operational cost of errors may be more useful.

Regression metrics

Mean absolute error (MAE) is the average absolute difference between predictions and actual values. Mean squared error (MSE) averages squared errors and penalizes large errors more strongly; root mean squared error (RMSE) is the square root of MSE and is expressed in the target's units. Metric choice depends on the problem and the consequences of error.

Thresholds and business trade-offs

Some classifiers produce scores or probabilities, then apply a threshold to assign a class. Lowering a threshold often increases recall but may also increase false positives. The appropriate threshold depends on costs, capacity, safety, and the intended use. Report the operating threshold with the metric.

Practical scenario

A medical screening model may prioritize recall to reduce missed cases, while a system that automatically blocks legitimate payments may need to limit false positives. Neither metric should be chosen in isolation from the context.

Checkpoint

Evaluate on unseen, representative data; use metrics suited to the task; examine the kinds of mistakes, not only one summary score.

Module 06: Overfitting, Underfitting & Generalisation

Training performance can be misleading. A useful model should learn patterns that continue to work on new examples rather than memorizing training records or failing to capture the task.

Learning objectives

- Diagnose overfitting and underfitting.
- Explain regularisation and cross-validation.
- Describe the bias–variance trade-off in plain language.

Overfitting

Overfitting occurs when a model fits quirks or noise in the training data and performs poorly on unseen examples. A common warning sign is very strong training performance paired with weaker validation performance. Possible responses include simplifying the model, adding suitable data, regularisation, early stopping, or better feature selection.

Underfitting

Underfitting occurs when a model is too simple or insufficiently trained to capture important patterns. It often performs poorly on both training and validation data. Potential remedies include a more expressive model, better features, reduced excessive regularization, or more appropriate training.

Regularisation

Regularisation discourages unnecessary complexity to improve generalization. L1 regularisation can encourage some coefficients to become zero; L2 regularisation penalizes large coefficients. Other techniques include dropout in neural networks and early stopping. Regularisation strength should be selected using validation data.

Cross-validation

In k-fold cross-validation, data is divided into k folds. The model is trained on k–1 folds and evaluated on the remaining fold, repeating until each fold has served as the held-out fold. Scores are summarized to estimate stability. Time-series and grouped data require split strategies that preserve time or group boundaries.

Bias–variance trade-off

A model with high bias is too restrictive and misses patterns; a model with high variance reacts too strongly to details in its training sample. Model development seeks a useful balance. More complexity is not automatically better; performance on suitable unseen data is the deciding evidence.

Practical scenario

A decision tree achieves 100% training accuracy but only 72% validation accuracy. This gap suggests overfitting. Pruning the tree, limiting its depth, using cross-validation, or gathering more representative data may help.

Checkpoint

Poor training and validation results suggest underfitting; excellent training results but weak validation results suggest overfitting.

Module 07: Core Algorithms in Practice

Different algorithms suit different target types, data structures, interpretability needs, and operating constraints. No algorithm is best for every task.

Learning objectives

- Describe linear and logistic regression.
- Explain decision trees and random forests.
- Match common algorithms to problem types.

Linear regression

Linear regression predicts a continuous numeric value using a weighted combination of input features. A simple form is ■ = wx + b, where x is an input, w is a learned weight, and b is a bias term. It is useful as an interpretable baseline for problems such as price or demand estimation when a linear approximation is reasonable.

Logistic regression

Despite its name, logistic regression is commonly used for classification. It maps a weighted combination of features through a logistic function to estimate a probability for a binary class. A decision threshold converts that probability into a predicted label. Its coefficients can provide useful insight, subject to feature scaling and model assumptions.

Decision tree

A decision tree makes predictions through a sequence of feature-based questions. Internal nodes split data; leaf nodes produce a class or numeric prediction. Trees are often easy to explain but can overfit if allowed to grow too complex. Depth limits and pruning can improve generalization.

Random forest

A random forest combines many decision trees trained with variation in samples and/or candidate features. Classification combines votes or probabilities; regression averages predictions. The ensemble often reduces the variance of a single tree, although it may be less interpretable than one small tree.

Choosing a starting algorithm

For a numeric target, linear regression can be a baseline. For binary classification, logistic regression or a tree-based model may be appropriate. For nonlinear tabular patterns, tree ensembles are often strong candidates. For images or audio, deep learning may be appropriate when sufficient data and compute are available. Always compare against a sensible baseline.

Practical scenario

A retailer wants to predict next week's sales amount: linear regression is a possible baseline because the target is numeric. The retailer wants to predict whether an order is fraudulent: logistic regression, a tree, or a forest could be compared as classification candidates.

Checkpoint

Identify the target first: continuous number or category. Then consider data type, baseline performance, interpretability, and error costs.

Module 08: Neural Networks Explained Simply

Neural networks learn adjustable transformations of input data. During training, the model's predictions are compared with targets, and its weights are updated to improve the chosen objective.

Learning objectives

- Describe neurons, layers, weights, and activation functions.
- Explain the forward pass and backpropagation conceptually.
- Identify when deep learning may justify its cost.

Neuron, weights, and layers

A neuron receives numeric inputs, multiplies them by weights, adds a bias, and applies an activation function. A network connects many such units into input, hidden, and output layers. Weights and biases are learned parameters; layer design determines how information is transformed.

Activation functions

An activation function introduces non-linearity so a network can represent more complex relationships than a stack of purely linear transformations. ReLU is common in hidden layers, while sigmoid or softmax may be used in output layers for particular classification setups. The correct output activation depends on the task and loss.

Forward pass

During a forward pass, input data moves through the network to produce a prediction. The prediction is compared with a target using a loss function. This calculates how poorly the model performed for that example or batch.

Backpropagation and parameter updates

Backpropagation uses the chain rule to compute gradients that indicate how changes in parameters affect the loss. An optimizer, such as stochastic gradient descent or Adam, uses these gradients to update weights and biases. Backpropagation calculates gradients; the optimizer performs the update.

Epoch and batch

An epoch is one complete pass through the training dataset. A batch is the group of examples processed in one step. Batch size and the number of epochs affect computation and learning behavior. More epochs do not always improve a model; training can overfit.

When deep learning is worth it

Deep learning can be powerful for complex unstructured data such as images, audio, and language. It may require more data, compute, time, and monitoring than simpler models. Begin with the problem and baseline; use a neural network when the expected performance or capability justifies the additional cost.

Practical scenario

A neural network is trained to classify fruit images. It makes predictions in the forward pass, computes classification loss, uses backpropagation to calculate gradients, and updates weights. After repeated batches and epochs, its performance is evaluated on held-out images.

Checkpoint

Forward pass produces predictions; loss measures error; backpropagation computes gradients; the optimizer updates parameters; an epoch is one pass through the training data.

Module 09: Responsible & Ethical AI

Responsible AI aims to make systems useful while addressing fairness, transparency, privacy, safety, accountability, and the consequences of errors.

Learning objectives

- Identify sources of algorithmic bias.
- Explain explainability and why it matters in consequential decisions.
- Describe human review, monitoring, and basic deployment safeguards.

Algorithmic bias

Algorithmic bias is systematic unfairness in outputs. It can arise from unrepresentative training data, biased historical decisions, flawed labels, poor feature choices, or a mismatch between the target and the real goal. Evaluate performance across relevant groups and investigate whether errors or outcomes differ in ways that create unfair harm.

Explainability

Explainability concerns how well people can understand why a model produced an output. Simple models may be more directly interpretable; other models may need explanation tools, feature-importance analysis, or examples. Explanations can be approximate and should not be treated as proof that a model is correct or fair.

Human-in-the-loop

A human-in-the-loop design places people in a review, approval, or override role. This can be important for medical, financial, employment, safety, or other high-impact decisions. Human review is effective only when reviewers have sufficient information, time, authority, and training; it should not become a meaningless rubber stamp.

Privacy and data governance

Collect only data needed for a legitimate purpose, protect access, define retention rules, and follow applicable privacy and security requirements. Sensitive or personal data should be handled carefully. Removing names alone may not prevent re-identification when other fields can be combined.

Model drift and monitoring

Model drift is a broad term for changes that can make a model less reliable after release. Input distributions, relationships between inputs and targets, user behavior, or operating conditions may change. Monitor data quality, input patterns, predictions, latency, errors, and—when labels become available—actual performance.

Practical safeguards

Document the intended use and known limitations; assess data representativeness; test subgroup performance where appropriate; secure the system; provide escalation and appeal paths; log important events; monitor live behavior; and define who can pause or roll back a model.

Practical scenario

A recruitment model trained on historical hiring decisions may reproduce past exclusion if those decisions or the data reflect bias. The team should inspect data and outcomes, test relevant groups, document limitations, retain human oversight, and monitor the system after release.

Checkpoint

Responsible AI is not just a model metric. Consider fairness, explainability, privacy, human accountability, and monitoring across the lifecycle.

Module 10: From Notebook to Deployment

Deployment turns a trained model into a service that an application can use. Reliable deployment includes versioning, input validation, monitoring, and a plan to respond when behavior changes.

Learning objectives

- Describe the lifecycle from experiment to monitored service.
- Explain how an API can expose predictions.
- Identify important post-release monitoring signals.

From experiment to service

A typical lifecycle is: define the problem; collect and prepare data; train and evaluate a model; package the model and preprocessing steps; register a version; deploy it to a service; monitor live behavior; and retrain or roll back when justified. Reproducibility requires tracking code, data versions, parameters, and evaluation results.

Model serving

Model serving exposes a trained model through an interface such as a REST API. A client sends a valid request, the service checks and transforms the input, runs inference, and returns a response. A production service should handle invalid requests, authentication, timeouts, failures, and sensible limits.

Inference

Inference is using a trained model to produce a prediction for new input. It differs from training, which fits model parameters. Inference may run in real time for an interactive application or in batches for large offline workloads.

Model registry

A model registry is a versioned store of models and their associated metadata, such as training details, metrics, approvals, and status. Versioning helps teams identify which model is live, compare releases, reproduce results, and roll back to a previous version when necessary.

Monitoring

Monitor service health such as latency, error rates, throughput, and availability; data quality such as missing or invalid fields; input and prediction distributions; and actual predictive quality when ground-truth labels arrive. Monitoring should include alerts and a defined response process, not merely a dashboard.

Release and rollback

Before release, validate the model artifact, preprocessing, API contract, security, and performance under expected load. Deploy gradually when possible. If errors, drift, or unacceptable outcomes appear, investigate, switch to a known-good version, or disable the model according to the system's risk plan.

Practical scenario

A web app sends a customer record to a prediction API. The API validates the fields, applies the same preprocessing used during training, runs inference, and returns a risk score. Logs track latency and failures; a registry records the model version; alerts notify the team if data patterns or service health change.

Checkpoint

Serving exposes the model; inference generates a prediction; a registry tracks versions; monitoring detects problems and supports a response.

AIML Quick Reference

Concept	Quick reminder
AI / ML / DL	AI is broad; ML is a subset of AI; DL is a subset of ML.
Supervised learning	Learns from labeled input–target examples.
Unsupervised learning	Finds structure in data without target labels.
Reinforcement learning	Agent learns from actions and rewards in an environment.
Feature / target	Feature is an input; target is what the model predicts.
Training / validation / test	Fit parameters / tune choices / final held-out evaluation.
Data leakage	Unavailable or held-out information contaminates training or evaluation.
Overfitting / underfitting	Overfits training details; underfits when model fails to capture patterns.
Regularisation / cross-validation	Control complexity / estimate performance across folds.
Linear / logistic regression	Linear regression predicts numeric values; logistic regression commonly predicts class probabilities.
Decision tree / random forest	Tree uses sequential splits; forest combines many trees.
Neuron / activation	Neuron combines weighted inputs; activation introduces non-linearity.
Backpropagation / epoch	Backpropagation computes gradients; epoch is one pass through training data.
Bias / explainability	Check systematic unfairness; make model behavior understandable enough for its use.
Serving / inference	Serving exposes a model; inference uses it to predict on new inputs.
Registry / monitoring	Registry tracks model versions; monitoring watches service, data, and quality.

General problem-solving checklist

- Define the task and the cost of mistakes before selecting a model.
- Check that the data is relevant, representative, clean, and legally/ethically usable.
- Separate training, validation, and test data to avoid leakage.
- Compare with a simple baseline and use task-appropriate metrics.
- Inspect errors and subgroup outcomes rather than relying on one score.
- Document assumptions, limitations, versions, and intended use.
- Monitor live performance and have a plan for escalation, rollback, or retraining.

Practice Assessment

Choose the best answer for each question. These questions cover the supplied AIML module descriptions and supporting foundational concepts. They are practice questions, not the live platform assessment.

1. Which statement best describes AI?

A. Every automated process

B. The broad field of systems performing tasks associated with human intelligence

C. Only robots

D. Only neural networks

2. A script sends a reminder at the same time each day using a fixed rule. This is primarily:

A. Reinforcement learning

B. Deep learning

C. Rule-based automation

D. Unsupervised learning

3. A dataset contains examples paired with known correct labels. Which paradigm fits?

A. Supervised learning

B. Unsupervised learning only

C. Reinforcement learning only

D. No learning method

4. A retailer groups customers without predefined labels. Which approach is a reasonable starting point?

A. Clustering

B. Fixed scheduling

C. Backpropagation as a business metric

D. API authentication

5. In reinforcement learning, an agent learns primarily through:

A. Actions and rewards from an environment

B. Only a held-out test label

C. A fixed report schedule

D. Data encoding alone

6. What is a feature?

A. A model version number only

B. An input variable used to make a prediction

C. Always the correct answer

D. The test score

7. What is the main purpose of a held-out test set?

A. To fit model weights repeatedly

B. To estimate final performance on unseen data

C. To tune every hyperparameter until the score is highest

D. To replace data cleaning

8. What is data leakage?

A. A model API returning JSON

B. A chart with missing colors

C. Information unavailable at prediction time contaminating training or evaluation

D. A model being stored in a registry

9. Training accuracy is excellent but validation performance is poor. This suggests:

A. Overfitting

B. Perfect generalization

C. Guaranteed fairness

D. No need for monitoring

10. Which algorithm commonly predicts a continuous numeric value?

A. Linear regression

B. A fixed timer

C. A model registry

D. An API gateway

11. Logistic regression is commonly used to:

A. Store model versions

B. Estimate class probabilities for classification

C. Replace all data preparation

D. Monitor latency only

12. A random forest generally combines:

A. Many decision trees

B. Only one handwritten rule

C. Multiple PDF files

D. API requests without a model

13. What does an activation function help a neural network do?

A. Introduce non-linearity

B. Guarantee zero errors

C. Prevent all bias

D. Replace the training dataset

14. What does backpropagation compute during neural-network training?

A. Gradients used to guide parameter updates

B. The legal consent of users

C. The final PDF page count

D. A deployment URL

15. Which statement about model bias is most accurate?

A. It can arise from unrepresentative data or flawed problem framing

B. It is always fixed by increasing model size

C. It is the same as API latency

D. It means every prediction is correct

16. What is the purpose of human-in-the-loop design?

A. Remove every human from a process

B. Let people review or approve model outputs where appropriate

C. Guarantee that models never drift

D. Make all models deep-learning models

17. Model serving means:

A. Exposing a trained model through an interface applications can call

B. Collecting training labels only

C. Calculating the number of epochs only

D. Deleting model versions

18. Inference is:

A. Fitting parameters from training data

B. Using a trained model to produce predictions on new input

C. Splitting a PDF

D. Defining an ethical policy only

19. Why is a model registry useful?

A. It tracks versions and associated metadata

B. It guarantees fairness automatically

C. It replaces testing

D. It prevents every possible error

20. Which item is useful to monitor after deployment?

A. Only the model's original training score

B. Latency, errors, data quality, input patterns, and predictive quality when labels arrive

C. Nothing once deployed

D. Only the color of the dashboard

ROYSTIAN FOUNDATION

Practice Assessment — Answer Key

Q	Answer	Explanation
1	B	AI includes perception, reasoning, language, prediction, and related tasks; it is broader than ML or deep learning.
2	C	A predefined schedule can automate a task without learning from data.
3	A	Supervised learning uses labeled input–target pairs.
4	A	Clustering is an unsupervised method for grouping similar records.
5	A	Reinforcement learning uses feedback from interaction with an environment.
6	B	Features are inputs; the target is the value being predicted.
7	B	The test set should be kept separate from training and model selection.
8	C	Leakage makes evaluation misleading and can cause production failure.
9	A	A large train–validation gap is a common sign of overfitting.
10	A	Linear regression is commonly used for numeric regression targets.
11	B	Logistic regression is a classification algorithm despite its name.
12	A	A random forest is an ensemble of decision trees.
13	A	Non-linear activations let networks model more complex relationships.
14	A	Backpropagation calculates how the loss changes with model parameters.
15	A	Bias can be introduced through data, labels, features, objectives, and design choices.
16	B	Human review can add accountability for consequential decisions when properly designed.
17	A	Serving makes the model accessible to an application, often through an API.
18	B	Inference uses a trained model; training is the fitting process.
19	A	Versioning supports traceability, reproducibility, and rollback.
20	B	Monitoring helps detect service problems and changing data or model performance.

Suggested self-check: 14/20 is 70%. Use the score as a revision guide only; the official pass result is determined by the platform's assessment.

Final readiness checklist

- I can explain the key vocabulary in all chapters.
- I can select a learning paradigm for a scenario.
- I can explain how data preparation and leakage affect evaluation.
- I can identify overfitting and underfitting from train/validation behavior.
- I can match common algorithm families to task types.
- I can explain the purpose of neural-network components and training steps.
- I can identify ethical safeguards and deployment monitoring needs.